

Slides 4.16: Careers in Housing, Interior Design, and Sustainability

Slide outline for Lesson 4.16. Thirteen slides, one day. Every pay figure on these slides is to be refreshed before teaching.

Slide 1: Do now

- Grandpa Okafor's room is going to be built for real: a wider door, a new outlet by the bed, fresh paint, and a walk-in shower next door
- Name three different people, by job, who would work on it
- One minute Image: the Okafor client card next to a photo of a room stripped to the studs. Notes: Take answers in a column on the board. Expect "builder" and "designer." Leave the column up; you add to it all period.

Slide 2: Thirteen people

- I counted thirteen
- Some of them make more than the architect
- Some of them started at sixteen in a high school shop class
- By the end of today you will know which one you would be Notes: Do not give the thirteen away yet. Add each one to the board column as it comes up in the chain.

Slide 3: The chain, part one: on paper

- Architect: draws it, checks that the walls can take it, stamps the plan
- Interior designer: plans the layout, the furniture, the color, the light, the finishes
- Sustainability consultant: picks the low-VOC paint and the LED lighting plan, files the rebate paperwork
- Landscape designer: the ramp at the right slope, the handrail, a clear path outside Image: a chain of four linked icons: a drafting pen, a paint chip fan, a leaf on a blueprint, a tree and a path. Notes: Students fill the Project Chain sheet as you go. Say out loud what the interior designer did: exactly what they did in Project 04, for pay.

Slide 4: The chain, part two: on site

- General contractor: runs the job, hires the trades, keeps the schedule and the budget
- Electrician: the new outlet, the switch moved to 42 inches, the night lights on the path
- Plumber: the shower drain and the walk-in shower with a grab bar
- HVAC technician: sizes the heat and air, vents the bathroom so it does not grow mold
- Furniture maker: the bed at 22 inches high, because no store sold one Image: a hard hat, a wire and switch, a wrench, a thermostat, a saw and chair.

Slide 5: The chain, part three: checking and running it

- Home inspector: checks the finished work against the code, for the buyer or the town
- Energy auditor: tests where the house leaks heat and says what to fix first
- Real estate agent: sold the family this house, and will list it someday

- Property manager: would approve this work if the family rented Notes: The contractor builds. The inspector checks the builder and works for the buyer or the town, not for the builder. Students confuse these two every year.

Slide 6: Three ways in

- College degree: architect, most interior designers, landscape architect, sustainability consultant
- Apprenticeship or trade school after high school CTE: electrician, plumber, HVAC, furniture maker, most contractors
- License or certification after a course and experience: real estate agent, home inspector, energy auditor, property manager
- An apprentice is paid from day one. Four to five years, about 8,000 hours, plus classes. Notes: “Apprenticeship” gets heard as “not a real job.” Say the pay out loud on the next slide before that idea sets.

Slide 7: The numbers

- Electrician: median about \$62,000
- Interior designer: median about \$62,000
- Plumber: about \$63,000. HVAC: about \$60,000.
- Architect: about \$95,000. Construction manager: about \$105,000.
- Furniture maker: about \$42,000. Real estate agent: about \$55,000, on commission.
- Union electricians in the New York area often earn more Image: a simple bar chart of the medians, sorted low to high, with the path labeled under each bar. Notes: The electrician and the interior designer earn about the same, and one of them pays for four years of school. Let that sit. Then say the numbers move every year and the cards say where each came from.

Slide 8: Sort the cards, 12 minutes

- First sort: by path. Degree, apprenticeship or trade, license after experience.
- Second sort: by where the person works most. An office, a job site, a client’s home, all three.
- Then pick three cards and fill the comparison table
- What they do, a day in the life, the path, the pay, one thing you would like, one you would not Notes: Circulate with the clipboard: path sort correct at the table, three careers compared with a path named for each. A student who writes “college” for the electrician gets the apprenticeship question before the self-check.

Slide 9: Questions I will ask

- Who wires the outlet? Who checks that it was wired right? Are those the same person?
- Which three careers here do not need a four-year degree? What do they need instead?
- What is an apprenticeship?
- An energy auditor did not exist fifty years ago. Which job here might not exist in fifty years, and which new one might? Notes: Take the fifty-years question with the whole class if there is time. Answers usually land on the auditor, the consultant, and something about remote work.

Slide 10: Would I like this?

- Twelve statements, rated 1 to 4
- Use what you actually did in Project 04, not what sounds good
- I liked measuring and drawing the plan. I liked building the model. I liked the budget. I liked pitching.
- Each line points to careers. Total by career. Image: the self-check sheet with the first four rows rated. Notes: Nine minutes. This is the lesson's High objective, so protect the time.

Slide 11: Two careers, with evidence

- Pick two careers and rate your fit 1 to 4
- Evidence comes from the project, not from a feeling
- "I rated electrician a 3 because I liked building the model, but I did not like the path measuring" Notes: Four students share a career and their evidence. Push anyone who says "it sounds cool" back to the project.

Slide 12: Where housing is going

- Two facts: more people aged 65 and over every year; households are smaller than fifty years ago (U.S. Census Bureau)
- Two technologies: sensors and controls (thermostat, leak sensor, lock on a phone); heat pumps, better insulation, and rooms built in a factory
- Pick one fact and one technology
- What will a house designed in 2045 have that yours does not? What will it not need any more?
- Every prediction on this board is somebody's job in ten years

Notes: Four minutes. Facts are stated as a direction, not a figure, so nothing on this slide goes stale. Take three predictions out loud and push each one to name its fact and its technology. Block the gadget answer with one question: "What has to change about the room for that to work?" This is where NYS EDM 2. e) is taught, and it is taught nowhere else in Unit 4, so it does not get cut for time. Image: two small line drawings side by side, the same room in plan view, one labeled "now" with a chimney and a narrow doorway, one labeled "2045" with a no-step entrance, a wide doorway, and a small mechanical closet.

Slide 13: The bridge

- A student who can draw a plan to scale, price a room, and pitch it has already done the first week of every one of these jobs
- The CTE programs near us that lead to these trades: the local BOCES or high school CTE partner
- Write one program name on your self-check
- Exit line: "The career I fit best is ___ because in the project I . **The path to it is .**" Notes: Two exit lines read aloud. Sort the slips by path. If a class chose the degree path only, open the Lesson 4.17 review with the trades bridge again. Keep the slips; Unit 7's career plan comes back to them.